Semiconductor Memory

1. The two tables below show a sample of commands available when interfacing to NAND flash (Table 1) and NOR flash (Table 2). What can you deduce from the nature of the Read/Program commands supported by the NAND and NOR Flash?

Table 1: NAND Flash Commands

Operation	Cycle 1	Cycle 2	Valid During Busy
PAGE READ	00h	30h	No
PAGE READ CACHE MODE START[1]	31h	–	No
PAGE READ CACHE MODE START LAST[1]	3Fh	–	No
READ for INTERNAL DATA MOVE[2]	00h	35h	No
RANDOM DATA READ[3]	05h	E0h	No
READ ID	90h	–	No
READ STATUS	70h	–	Yes
PROGRAM PAGE	80h	10h	No
PROGRAM PAGE CACHE[1]	80h	15h	No
PROGRAM for INTERNAL DATA MOVE[2]	85h	10h	No
RANDOM DATA INPUT for PROGRAM [4]	85h	–	No
BLOCK ERASE	60h	D0h	No
RESET	FFh	–	Yes

Table 2: NOR Flash Commands

		WRITE Operations											
		1st		2nd		3rd		4th		5th		6th	
Command	Length	Addr	Data	Addr	Data	Addr	Data	Addr	Data	Addr	Data	Addr	Data
READ/RESET	1	X	F0										
	3	555	AA	2AA	55	X	F0						
AUTO SELECT	3	555	AA	2AA	55	555	90						
PROGRAM	4	555	AA	2AA	55	555	A0	PA	PD				
UNLOCK BYPASS	3	555	AA	2AA	55	555	20						
UNLOCK BYPASS PROGRAM	2	X	A0	PA	PD								
UNLOCK BYPASS RESET	2	X	90	X	00								
CHIP ERASE	6	555	AA	2AA	55	555	80	555	AA	2AA	55	555	10
BLOCK ERASE	6+	555	AA	2AA	55	555	80	555	AA	2AA	55	BA	30
ERASE SUSPEND	1	X	B0										
ERASE RESUME	1	X	30										
READ CFI QUERY	1	55	98										

2. The SSD controller manage the translation between Logical Block Address (LBA) from the Application processor to the Physical Block Address (PBA) of the actual storage location of the data in the SSD.
    - Table 3.1 shows the mapping of LBA to PBA.
    - For simplicity's sake, we can assume that these addresses at block level address, e.g. LBA block 0 (LBA0) is mapped to PBA block 9 (PBA9) in the table.
    - Table 3.2 shows the erase count of each PBA block.
    - LBA, PBA block correspond to the flash block in the flash structure, i.e. erasure is at block level.
    - A 'Y' in the 'Empty' column indicates that the Flash Block is empty.

(a) Complete Table 3.1 for Dynamic wear leveling algorithm. You can further assume that
    - Access sequence from the processor is LBA 7, 4, 3, A.
    - All access will go through the wear levelling algorithm for physical block assignment.
    - If the LBA was originally assigned to a different PBA, the PBA containing the old data will be erased.

    What is the consideration to take for the erasure process of the old data?

Table 3.1

LBA	PBA
0	9
1	X
2	X
3	5
4	X
5	4
6	6
7	7
8	X
9	X
A	0

Table 3.2

Erase Count Table		
PBA	Erase Count	Empty
0	5	N
1	33	Y
2	15	Y
3	98	Y
4	45	N
5	11	N
6	69	N
7	78	N
8	A3	Y
9	F1	N
A	56	Y

(b) In table 3.2, you can see that PBA block 6 has a very low erase count but would not be evaluated under Dynamic wear levelling algorithm. Static wear levelling attempts to relocate such flash blocks that are seldom re-written to achieve more even erase count distribution. There are many ways to implement Static wear levelling, below is one example. Go through the algorithm described and suggest enhancement to the algorithm.
    - On every access, all blocks (occupied and empty) will be evaluated for block assignment.

- PBA block with the lowest erase count will be selected to store the data.

3. Figure 5.2a and 5.2b shows the Read and Write timing diagram of a processor's external memory interface. The processor can configure the access timing via three parameters: setup, strobe and hold. The description of these parameters can be found in Figure 5.1. The example in the figure is configured for setup = 2, strobe = 5 and hold =1. Figure 5.3a and 5.3b shows the read and write access timing diagram of a SRAM. Configure the processor's external memory interface to match SRAM timing. Following are some information that you'll need for the computation
   - CPU Clock cycle = 5 ns
   - The unit of the setup-strobe-hold parameters are in CPU clock cycles.
   - Processor ARE_ pin is connected to SRAM OE_ pin
   - Processor latch in the data on rising edge of ARE_
   - Maximum value of SRAM tOE = 40 ns
     - tOE is the Output Enable to Output Valid timing.
   - Minimum value of SRAM tWP = 40ns
     - tWP is the Write Pulse Width timing
   - Minimum tDH = 5 ns
     - tDH is the Data Hold from WE signal inactive
   - Minimum tAS = 0 ns.
     - tAS is the Address setup time before WE signal is active

Parameters	Control Bits	Definition
Setup periods	RDSETUP WRSETUP	A setup period is the time in CPU clock cycles given to setup the address, chip enable ($\overline{CE}$), and byte enable ($\overline{BE}$) signals before the read strobe signal ($\overline{ARE}$) or write strobe signal ($\overline{AWE}$) falls. For an asynchronous read operation, this is also the setup period for the output enable signal ($\overline{AOE}$) before $\overline{ARE}$ falls.
Strobe periods	RDSTROBE WRSTROBE	A strobe period is the time in CPU clock cycles between the falling (activation) and rising (deactivation) of the read or write strobe signal.
Hold periods	RDHOLD WRHOLD	A hold period is the time in CPU clock cycles during which the address and byte enable lines are held active after the read or write strobe signal rises. For an asynchronous read operation, this is also the hold period for the output enable signal after $\overline{ARE}$ rises.

Figure 5.1: External Memory Interface Configuration Parameters

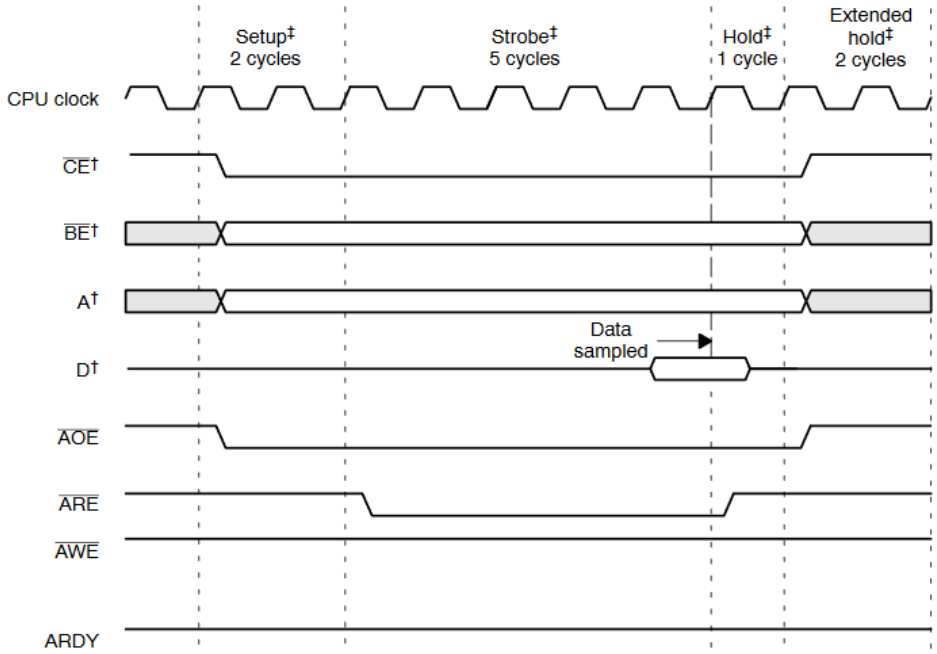


Figure 5.2: Processor External Read Timing Diagram

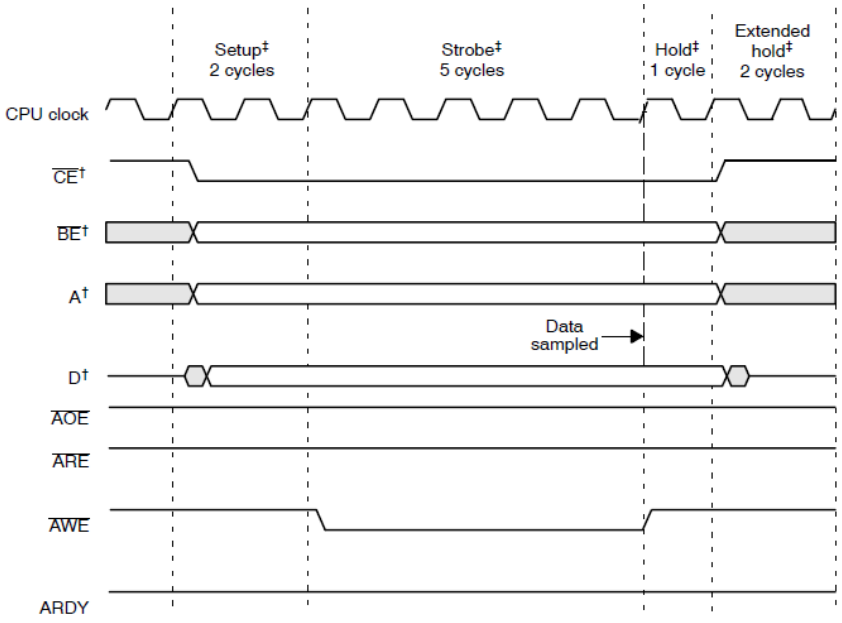


Figure 5.2b: Processor External Write Timing Diagram

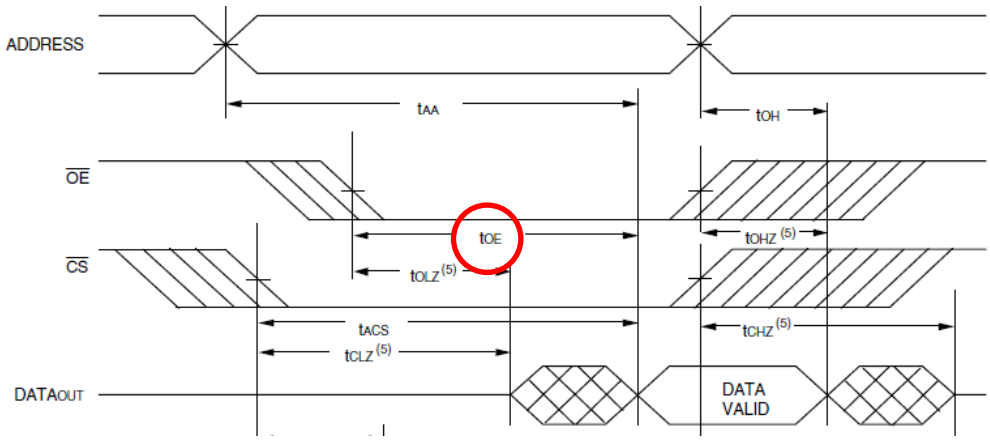


Figure 5.3: SRAM Read access timing diagram

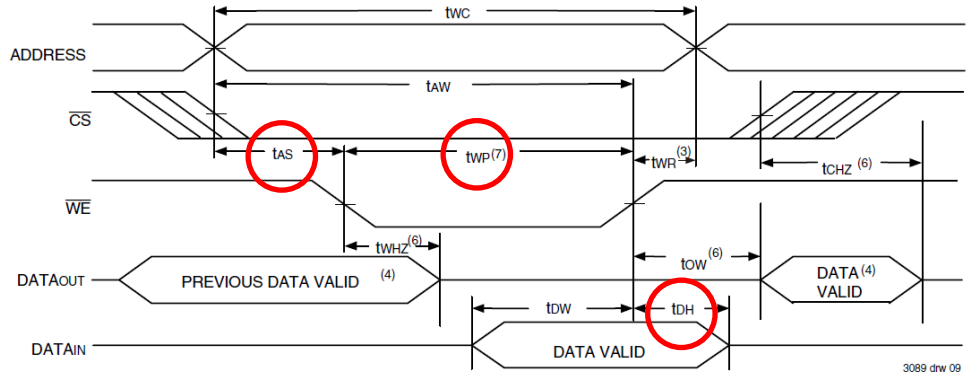


Figure 5.3b: SRAM Write access timing diagram

4. Complete the timing diagrams in Table 6.1 and 6.2 below, fill in the missing information on commands, data, addresses etc.
   - What is the CAS latency of the SDRAM for Table 6.1?

Table 6.1: SDRAM Read without Auto-Precharge

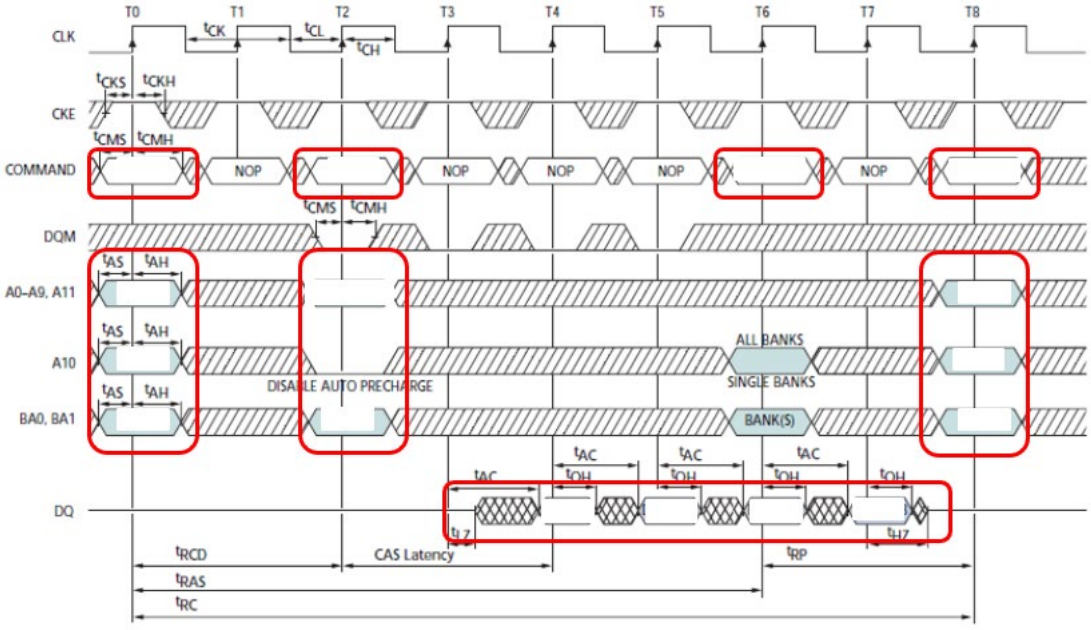
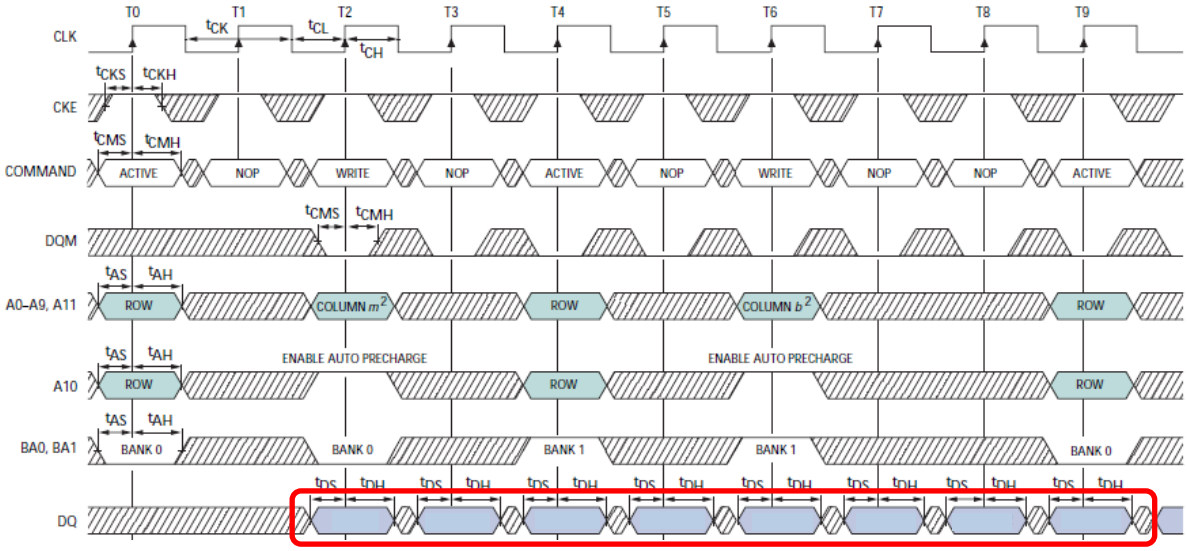


Table 6.2: SDRAM Write – Alternating Bank Access (with Auto-Precharge)



5. Arrange the access sequence to optimise the access time for the following data. Given that the tRCD(min) = 2 SDRAM clocks. tRCD is the minimum delay needed after an ACTIVE command before user can issue the READ/WRITE command to the SDRAM. Hint: Check the lecture notes for similar discussion.

    - Bank1-Row2-Byte0 (1 byte)
    - Bank3-Row3-Byte2 (3 bytes)
    - Bank1-Row3-Byte2 (3 bytes)
    - Bank1-Row2-Byte3 (1 byte)

(Optional)

6. Description and explanation of various SSD Design considerations: homework given during lecture